Angelina Ju

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Education

University of Tennessee, Knoxville *Knoxville, TN*
Bachelor of Science in Computer Science *Expected May 2027*
GPA: 3.79/4.00 – Dean's List

Research Experience

Undergraduate Research Assistant *Knoxville, TN*
Global Computing Laboratory, EPIC Lab, University of Tennessee, Knoxville *Aug. 2025 – Present*

- Developed the JupyterLab interface for PegasusAI, an in-notebook AI assistant deployed in a data engineering course, embedding AI help in students' coding environment so they can ask for help without leaving their workflow.
- Load-tested the assistant's model serving under varying numbers of concurrent users to measure throughput and scalability, ensuring responsive human–AI interaction at classroom scale.
- Co-authored two manuscripts on the deployment: one characterizing students' help-seeking styles, trust, and perceived helpfulness, and one formalizing human–AI interactions as structured, analyzable data.

Undergraduate Research Assistant *Knoxville, TN*
EPIC Lab, Zhao Robotics Lab, University of Tennessee, Knoxville *Mar. 2024 – Apr. 2025*

- Explored how quadruped robots could support physical activity and companionship for older adults with dementia, using ROS and Python for simulation and robot control.
- Collaborated with three interdisciplinary teams on human-robot interaction, mobility, and safety challenges, and helped plan a participatory design session with older adults (surveys, robot demonstration, group discussion, and collage-based ideation).
- Designed and refined conversational AI prompts to make the robot's conversational responses clearer and more natural for users.

Publications

- Taufer, M., Ng, K.W., Lumsden, I., Pandey, A., Gore, R., **Ju, A.**, Moudgalya, S., Deelman, E., Swaminathan, S. Human–AI Interactions as First-Class Data: A Formal Interaction Model and Runtime for Explainable AI Collaboration. *IEEE International Conference on Big Data (BigData 2026)*. Under review.
- Moudgalya, S., Ng, K.W., Lumsden, I., **Ju, A.**, Gore, R., Deelman, E., Taufer, M., Swaminathan, S. Who Uses the AI Assistant, and Why? Engagement Profiles and Help-Seeking Styles in a Data Engineering Course. In preparation.

Presentations and Talks

Walking the Dog: Promoting Mobility and Emotional Support through Assistive Quadrupedal Robots for Dementia Care (Poster)

- **Ju, A.**, Dy, A., Scott, D., Morris, T., Reig, S., Chidziwisano, H., Moudgalya, S., Zhao, X., Swaminathan, S.
- Exhibition of Undergraduate Research and Creative Achievement (EURēCA), Knoxville, TN, Apr. 2025.

Evolutionary Optimization of Bridge Design Using Genetic Algorithms (Poster)

- Dy, A., **Ju, A.**, Murray, C.
- COSC 420 Class Poster Session, University of Tennessee, Knoxville, TN, Spring 2026.

Grants

Faculty Research Assistants Funding (FRAF), University of Tennessee, Knoxville 2024

Honors and Awards

Society of Women Engineers (SWE) **Deloitte Scholarship** 2025, 2026

Tickle College of Engineering Differential Fee Scholarship, University of Tennessee, Knoxville 2026

Tau Beta Pi Engineering Honors Society, Tennessee Alpha Chapter 2026

Professional Experience

Product Management Intern *Memphis, TN*

St. Jude Children's Research Hospital, Research Support Services Team *June 2026 – August 2026*

- Researching integration of AI products for streamlining scrum ceremonies.
- Coordinating parallel UI/UX, retrieval-augmented generation (RAG), and Microsoft Teams integration workstreams so the assistant meets staff inside the tool they already use every day.
- Coordinating a user guide for ARIA Guide, a pediatric oncology knowledge platform serving 1,500+ healthcare providers in low- and middle-income countries, to make clinical information easier to find and use.

Product Engineering Intern *Remote*

Wiillow *June 2025 – July 2025*

- Worked with product and UX teams on an AI-powered social-emotional learning (SEL) chatbot for children, reaching over 80 users.
- Gathered requirements and planned features around children's needs, balancing user experience against technical feasibility and timeline constraints.
- Built responsive, child-friendly chat interfaces in React, TypeScript, and TailwindCSS, and supported backend features for emotional tone analysis and language development tracking.

Projects

Structural Evolution: Genetic Algorithms for Bridge Design *Spring 2026*

COSC 420 course research project, team of 3 – Python, headless physics simulator

- Evolved bridge designs with a genetic algorithm in a headless physics simulator, using a fitness function that rewarded vehicle progress and penalized material cost and time.
- Found that optimizing human-designed starting bridges was far more stable than evolving from scratch, cutting the most complex level's cost from $91,119 to $27,747, which suggests value in pairing human design with algorithmic optimization.

Quadrupedal Robot Reinforcement Learning *Mar. 2025 – May 2025*

PyBullet, PyTorch, Gym, NumPy

- Trained a Proximal Policy Optimization (PPO) policy from scratch so a quadrupedal robot could autonomously perform socially oriented behaviors, such as turning toward and following a target.
- Defined evaluation metrics and success criteria for these social behaviors and iterated on testing data, improving reward performance by 164% and reducing training time by up to 40%.
- Directed a team of 3 in refining the algorithm, and presented findings and design decisions to research faculty.

Fitness Tracker (Full-Stack Product) *Sep. 2025 – Dec. 2025*

React, TypeScript, FastAPI, PostgreSQL

- Scoped 5 core features (analytics dashboard, nutrition, exercise, and goal tracking, and user settings) around how users monitor their health, within a 3-month development cycle.
- Led a team of 5 developers, assigning work by skill set to deliver on schedule across sprints.

Leadership & Service

Product Manager *Knoxville, TN*

Hack4Impact UTK, student organization building software for nonprofits *July 2026 – Present*

- Leading product development for a nonprofit client: eliciting the organization's needs, translating them into

requirements, and coordinating a team of student developers to build around them.

Vice President *Knoxville, TN*
Systers: Women in EECS, University of Tennessee, Knoxville *August 2025 – Present*

- Plan events that build community and support for women in computing in the EECS department.
- Manage over $10,000 in annual funding to send students to the Grace Hopper Celebration of Women in Computing.

Conferences Director *Knoxville, TN*
Society of Asian Scientists and Engineers (SASE), University of Tennessee Chapter *May 2024 – Present*

- Lead end-to-end planning for the chapter's trip to the SASE National Convention, defining scope, timeline, and budget for 30+ students.
- Communicate across students, faculty, department heads, and sponsors, securing over $35,000 in funding.
- Delegate tasks to a planning committee, track progress, and resolve unforeseen issues in real time.

Coursework

Data Structures and Algorithms, Operating Systems, Introduction to Machine Learning, Introduction to Cybersecurity, Discrete Structures, Linear Algebra, Probability and Random Variables, Human-Computer Interaction, Software Engineering, Systems Programming, Operating Systems, Biologically-Inspired Computation, Introduction to Artificial Intelligence

Technologies

Programming Languages: Python, C/C++, C#, TypeScript, JavaScript, SQL (PostgreSQL), HTML/CSS

Frameworks & Libraries: React, TailwindCSS, FastAPI, Django, Node.js, PyTorch, NumPy, Pandas, PyBullet, Gym, ROS

Tools: Git/GitHub, Docker, Linux, VS Code, JupyterLab, Azure DevOps, Jira, Trello, Android Studio, Arduino, Microsoft 365

Product & Project Management: Agile/Scrum, Sprint Facilitation, Roadmapping, User Research, MVP Scope Definition, Requirements Documentation, Feature Prioritization, A/B Testing, Stakeholder Management

Languages: English (Native), Mandarin (Conversational)